Essentials of Web Programming

B. Tech (AI) III Year I Semester					Dept. of Artificial Intelligence			
Code	Category	Hours / Week			Credits	Marks		
		L	T	P	C	CIE	SEE	Total
EMA3181	Theoretical & Practical (PC)	3	0	2	4	50	50	100

Course Outcomes

After completion of course, students would be able to:

1. Design static web pages using HTML.
2. Develop responsive layouts using CSS and implement dynamic client-side scripts using JAVA Script.
3. Make use of XML and AJAX for structured data exchange and asynchronous communication.
4. Construct robust web applications using TypeScript.
5. Create single-page web applications using Angular.

UNIT-I

Introduction to HTML: Introduction to Web, Evolution of HTML, WWW, HTML Basics: Elements, Tags, Attributes, Headings, Paragraphs, Links, Lists, Tables, Forms, Semantic Elements, and tags, Multimedia Elements, HTML5.

UNIT-II

CSS: Introduction to CSS, Types of Style Sheets, Selectors, Pseudo Classes and Pseudo Elements, Colors, Fonts, Lists, Backgrounds, Box Model, Conflict Resolution.

JavaScript: Introduction to JavaScript, Need of JavaScript, Variables, Data Types, Operators, Conditional Statements, Loops, Functions, Arrow Functions, Scope, Arrays, Objects, DOM, Events.

UNIT-III

Advanced JavaScript: Classes, Inheritance, Modules, Asynchronous Programming, Callbacks, Promises, Async/Await, Fetch API, Form Validation.
XML: Syntax, Namespaces, XML Schema, XML Parsers.
AJAX: Introduction, Working Principle, XMLHttpRequest, JSON Response Handling.

UNIT-IV

TypeScript: Introduction to TypeScript, Installation, Variables, Data Types, Functions, Optional Parameters, Default Parameters, Rest Parameters, Arrow Functions, Interfaces, Duck Typing, Classes, Constructors, Access Modifiers, Namespaces, Modules, Generics, Generic Constraints.

UNIT-V

Angular: Introduction to Angular, Features, Angular CLI, Project Setup, Components, Modules, Templates, Data Binding, Directives, Pipes, Services, Dependency Injection, Lifecycle Hooks, HTTP Client, Template Driven Forms, Reactive Forms, Validation, Routing.

Text Books

1. Programming the World Wide Web, 7th Edition, Robert W Sebesta, Pearson, 2014.
2. Uttam K. Roy, Web Technologies, 8th Impression, Oxford Publication, 2014.
3. Simon Holmes, Clive Harber “Getting MEAN with Mongo, Express, Angular and Node”, Second Edition, Manning Publications Co., 2019.
4. Full Stack JavaScript Development with MEAN, Colin J Ihrig, Adam Bretz, 1st edition, SitePoint Pty. Ltd., O'Reilly Media.

Reference Books

1. Thomas Powell, “The Complete Reference HTML and CSS”, 5th Edition, Tata McGraw Hill, 2010.
2. Thomas Powell, Fritz Schneider, “The Complete Reference JavaScript 2.0”,3rd Edition, Tata McGraw Hill, 2012.
3. https://infyspringboard.onwingspan.com/en/app/toc/lex_18109698366332810000_shared/overview (Javascript)
4. https://infyspringboard.onwingspan.com/en/app/toc/lex_9436233116512678000_shared/overview (Typescript)
5. https://infyspringboard.onwingspan.com/en/app/toc/lex_20858515543254600000_shared/overview (Angular)